Sets

A **set** is a collection of objects. The objects do not have an order and they are not repeated. The objects are also called **members** or **elements** in the set. One way to specify a set is to list its members. So, for instance $\{2,4,6,8\}$ is the set of positive integers that are less than 10. Another way to specify this set is to do it using a set of rules or conditions. For example,

$$O = \{x \mid x \text{ is an even positive integer less than } 10\}$$

This is read as, "O is the set of elements x such that x is an even positive integer less than 10."

The notation $a \in A$ means that the object named "a" is an element of the set named "A." One reads this expression as, "a is an element of A." Similarly, the notation $a \notin A$ means that "a" is not an element of "A."

There are a number of important sets of numbers. Among them are

$\mathbb{N} = \{0,1,2,3,\ldots\}$, the set of **natural numbers**
$\mathbb{Z} = \{\ldots,-2,-1,0,1,2,\ldots\}$, the set of **integers**
$\mathbb{Z}^+ = \{1,2,3,\ldots\}$, the set of **positive integers**
$\mathbb{Q} = \{p/q \mid p \in \mathbb{Z},\ q \in \mathbb{Z},\ q \neq 0\}$, the set of **rational numbers**
$\mathbb{R}$ **,** the set of **real numbers**
$\mathbb{R}^+$, the set of **positive real numbers**
$\mathbb{C}$, the set of **complex numbers**

Specified as sets, intervals of real numbers are for $a, b \in \mathbb{R}$ and $a < b$

$[a,b] = \{x \mid a \leq x \leq b\}$
$[a,b) = \{x \mid a \leq x < b\}$
$(a,b] = \{x \mid a < x \leq b\}$
$(a,b) = \{x \mid a < x < b\}$

Two sets are said to be **equal** if they have the same elements. For instance, if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$.

The **empty set** is the set that contains no elements. The empty set is denoted as $\varnothing$ or $\{\}$. A set with a single element is called a **singleton**. For instance, $\{0\}$ is a set with a single element. Note that

$0 \neq \{0\}$
$0 \neq \varnothing$
$\{0\} \neq \varnothing$

The reason that $0 \neq \{0\}$ is that the left-hand side of the expression is a number and the right hand side of the expression is a set. There is no concept of equality between sets and numbers. For a similar reason, zero is not equal to the empty set, $0 \neq \varnothing$. In the expression $\{0\} \neq \varnothing$ both sides of the equation are sets. But remember that set equality means that the sets have the same elements. The left-hand side of the expression is a set that contains the element 0. However, the right hand side is the set that contains no elements. Since $0 \notin \varnothing$, the two sets are not equal.

The set A is a **subset** of B if and only if each element in A is also in B. This is written as $A \subseteq B$, and more formally means that

$$\forall x(x \in A \rightarrow x \in B)$$

To show that $A \subseteq B$ one needs to show that if $x \in A$ then $x \in B$. To show that $A \not\subseteq B$, find an $x \in A$ such that $x \notin B$.

Set Operations

If A and B are sets the **union** of A and B, written $A \cup B$, is the set of elements that are in A or B, or in both sets. Formally,

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

The **intersection** of sets A and B, written as $A \cap B$, is the set of elements that in both A and B. Said otherwise

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

Two sets are **disjointed** if they do not contain any identical elements, or if their intersection is the null set.

For sets, A and B, the **difference** of A and B, which is written $A - B$, is the set containing elements in A, but not in B. Formally the difference is

$$A - B = \{x \mid x \in A \wedge x \notin B\}$$

Often one can specify a **universal set** U. Once this is done, one then defines the **complement** of a set, A, to be those elements in U that are not in A. The complement of A is written as $\overline{A}$. It is specified as

$$\overline{A} = \{x \in U \mid x \notin A\}$$

Set Identities

The set identities, in the table below, follow from the logical equivalences studied earlier.

Set Identities

Identity	Name
$A \cap U = A$ $A \cup \varnothing = A$	*Identity Laws*
$A \cup U = U$ $A \cap \varnothing = \varnothing$	*Domination Laws*
$A \cup A = A$ $A \cap A = A$	*Idempotent Laws*
$\overline{(\overline{A})} = A$	*Complementation Laws*
$A \cup B = B \cup A$ $A \cap B = B \cap A$	*Commutative Laws*
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cup C$	*Associative Laws*
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	*Distributive Laws*
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	*De Morgan's Laws*
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	*Absorption Laws*
$A \cup \overline{A} = U$ $A \cap \overline{A} = \varnothing$	*Complement Laws*

(Adapted from Rosen, K.H. (2012) Discrete Mathematics and Its Applications. McGraw-Hill)

There is a notation, similar to the summation notation that you might have seen in earlier mathematics courses, that allows us to speak about the union and intersection of more than two sets. Suppose there is a collection of indexed sets called $A_1, A_2, \ldots, A_n$. The union of these sets is written as

$$A_1 \cup A_2 \cup \ldots \cup A_n = \bigcup_{i=1}^{n} A_i$$

The intersection of these sets is written as

$$A_1 \cap A_2 \cap \ldots \cap A_n = \bigcap_{i=1}^{n} A_i$$

Note: the interested student with a Calculus background might consider the union or intersection of an infinite number of sets.

Functions

If A and B are nonempty sets, then a **function** f from A to B is an assignment of exactly on element of B for each element in A. This is written as $f : A \to B$. For elements $a \in A$ and $b \in B$ one writes

$$f(a) = b$$

One says that A is the **domain** of f and B is the **codomain** of f. Further one says that b is the **image** of a and that a is the **preimage** of b. The set of all images for all elements in A is the **range** or **image** of f.

If the image of two functions f_1 and f_2 are real numbers, then arithmetic can be performed on the functions. Let f_1 and f_2 be functions from A to $\mathbb{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbb{R}$. For all $x \in A$

$$\begin{aligned} (f_1 + f_2)(x) &= f_1(x) + f_2(x) \\ (f_1 f_2)(x) &= f_1(x) f_2(x) \end{aligned}$$

A one-to-one function is a function that has a different image for each element in its domain. More precisely, a function f is said to be **one-to-one**, or an **injunction**, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f.

An onto function is a function where for every element in its range has a preimage. More precisely, a function f from A to B is called **onto**, or a **surjection**, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.

A function that is both one-to-one and onto is said to be a **bijection,** or a **one-to-one correspondence**.

A function whose domain and range are subsets of the real number is said to be an **increasing function** if $f(x) \leq f(y)$ whenever $x < y$. The function is said to be **a strictly increasing function** if the less than or equal to relation is a strictly less than relation i.e., $f(x) < f(y)$ whenever $x < y$. A **decreasing**

function satisfies the condition $f(x) \geq f(y)$ whenever $x < y$. Similarly, a **strictly decreasing function** satisfies the condition $f(x) > f(y)$ whenever $x < y$.

Suppose that f is a one-to-one correspondence that maps from set A to set B. The **inverse function of** f is the function that assigns to an element b in B a unique element a in A such that $f(a) = b$. The inverse function is denoted f^{-1} and one writes $f^{-1}(b) = a$.

The function f must be a one-to-one correspondence for its inverse to exist.

Suppose that g is a function from set A to B and that f is a function from set B to C. The **composition** of the functions f and g is defined for all $a \in A$ by

$$(f \circ g)(a) = f(g(a))$$

Function composition is not commutative, which means that generally $f \circ g$ and $g \circ f$ are not equal.

Sequences and Summations

A **sequence** is a function from a subset of the integers to a set S. The notation a_n denotes the image of the integer n. One calls a_n a **term** in the sequence.

For example, $a_n = 1/n$ for $n = 1, 2, 3, \ldots$ is the sequence $1, \frac{1}{2}, \frac{1}{3}, \ldots$.

A **geometric progression** is a sequence of the form

$$a, ar, ar^2, \ldots, ar^n, \ldots$$

where a and r are real numbers. For example, suppose that $a = 2$ and $r = -1/10$. This geometric progression is

$$2, \frac{-2}{10}, \frac{2}{100}, \frac{-2}{1000}, \ldots$$

An **arithmetic progression** is a sequence of the form

$$a, a + d, a + 2d, \ldots, a + nd, \ldots$$

where a and d are real numbers. For example, suppose that $a = -1$ and $d = 2$. This arithmetic progression is

$$-1, 1, 3, 5, \ldots$$

A sequence can also be defined by a recurrence relation. This means that a_n would be defined using prior terms in the sequence, like $a_{n-1}, a_{n-2}, \ldots$. The Fibonacci sequence is one example of a sequence that is specified by a recurrence relation.

The Fibonacci sequence $f_0, f_1, f_2, \ldots$ is defined by the initial conditions $f_0 = 1$, $f_1 = 1$, and the recurrence relation

$$f_n = f_{n-1} + f_{n-2}$$

for $n = 2, 3, 4, \ldots$. The first few terms of the Fibonacci sequence are $1, 1, 2, 3, 5, 8, 13, \ldots$. One sees that after the first two terms, a term is the sum of the prior two terms.